

Flood Risk Assessment of Lagos Coastal Road Corridor (2 km Buffer)

1. Overview

A geospatial flood risk assessment was conducted along the Lagos Coastal Road corridor using a **2 km buffer zone** to define the study extent. The analysis focused on identifying areas susceptible to flooding based on terrain elevation and slope characteristics within the corridor.

2. Data and Methodology

The assessment utilized the following datasets:

- Digital Elevation Model (DEM) for terrain analysis
- Coastal road alignment data
- Derived slope and hillshade layers for terrain characterization

The methodology involved:

- Clipping the DEM to the 2 km study corridor
 - Classifying elevation into flood risk categories:
 - **High Risk:** ≤ 2 m
 - **Medium Risk:** 2 – 5 m
 - **Low Risk:** > 5 m
 - Converting classified raster outputs into vector polygons
 - Calculating area coverage for each risk category
 - Supporting analysis using slope data to identify low-gradient zones prone to water accumulation
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3. Results

The total study area within the 2 km corridor is approximately **299 km²**.

Flood Risk Distribution

Risk Category	Area (km ²)	Percentage (%)
High Risk (≤ 2 m)	20.79	6.95%
Medium Risk (2–5 m)	63.81	21.34%
Low Risk (> 5 m)	214.39	71.71%

4. Key Findings

- The majority of the corridor (~**72%**) lies within relatively low flood risk zones.
 - However, **approximately 28%** of the study area falls within **moderate to high flood risk zones**, indicating significant localized vulnerability.
 - High-risk zones are primarily associated with **low-lying coastal and lagoon-adjacent areas**.
 - Slope analysis indicates that many of these vulnerable zones are characterized by **very low gradients**, which reduces surface runoff and increases the likelihood of water accumulation.
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5. Interpretation

Although large portions of the corridor are above critical elevation thresholds, flooding risk is **not uniformly distributed**. Instead, it is concentrated in **specific low-elevation, flat terrain pockets**, particularly along the Lekki–Eleko axis.

It is important to note that this assessment is based on terrain analysis. Actual flood occurrence in Lagos is influenced by additional factors such as:

- Intense rainfall events
 - Inadequate drainage infrastructure
 - Urban development and land reclamation
 - Tidal and coastal interactions
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6. Conclusion

The Lagos Coastal Road corridor exhibits **moderate overall flood exposure**, with critical high-risk zones that require targeted mitigation. While the general terrain suggests relatively low

vulnerability across most areas, the presence of **localized high-risk sections** highlights the need for:

- Improved drainage planning
 - Flood-resilient infrastructure design
 - Continuous monitoring of coastal and low-lying zones
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7. Recommendation (Optional but Strong)

- Prioritize engineering interventions in identified **high-risk segments**
 - Integrate hydrological modeling in future studies for detailed flood simulation
 - Enforce development controls in low-lying coastal zones
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 **You're done.**

This is:

- ✓ Structured
- ✓ Quantified
- ✓ Defensible
- ✓ Report-ready